IS507: Heart Failure Death Prediction Project

Summary:
Heart failure is a serious medical condition associated with substantial mortality. This project investigates whether clinical characteristics can predict death during follow-up among patients diagnosed with heart failure.

Using the UCI Heart Failure Clinical Records dataset (299 real patients, 12 clinical features, follow-up of 4 to 285 days) and additional artificial records generated from training data only, we will develop and compare Logistic Regression, Random Forest, and XGBoost models to predict death events. The models will be evaluated with repeated stratified cross-validation and a final held-out test set of real patients, using accuracy, precision, recall, F1-score, and ROC-AUC. A majority-class baseline (about 68% accuracy) is used for comparison, and 80% accuracy is kept as the target goal. Because follow-up time is not known when a prediction would be made and is tied to when death occurs, every model is evaluated both with and without it.

The project will determine which model performs best while considering the dataset's small original sample size and the limitations of using artificially generated records.

Research Problem:
Heart failure is a serious medical condition associated with substantial mortality. Identifying patients who are at higher risk of death could help clinicians better understand patient risk and potentially support earlier intervention. This project investigates whether clinical characteristics can be used to predict death during follow-up (up to about 10 months) among patients diagnosed with heart failure.

The type of statistical task:
This is a binary classification task. Using Logistic Regression, Random Forest, and XGBoost, we will predict whether a patient with diagnosed heart failure dies during the follow-up period. Follow-up in this dataset ranges from 4 to 285 days, so it contains no 1-, 2-, or 3-year outcomes, and survival over those horizons is outside the scope of this project. We will compare the three trained and tested models and identify the best one. The target is 80% accuracy, reported alongside recall, F1-score, and ROC-AUC.

How the data were generated or observed:
The original dataset consists of 299 patient records collected as part of a clinical study on patients with heart failure (Chicco & Jurman, 2020). The real data will be split first with a stratified 80/20 split (239 training patients, 60 test patients). An additional 1,000 artificial records will then be generated using **[generation method, e.g., SDV or CTGAN]** from the training data only, so that no information from test patients enters the artificial data. Inside cross-validation, artificial records will be generated from each training fold only. Artificial records will never be used for validation or testing, and care will be taken to distinguish original observations from artificial ones. Results will be reported for models trained on real data only and for models trained on real plus artificial data, both evaluated on the same held-out real patients.

The target population and observed sample:
The target population is patients diagnosed with heart failure who have clinical information that can be used to predict their risk of death. The observed sample consists of the 299 patients in the original UCI Heart Failure Clinical Records dataset. The dataset contains demographic, medical, and blood-test information for each patient. The additional 1,000 artificial records are intended to expand the available training data but may not represent independent real-world patients. They are never part of any evaluation.

Treatment of duplicated, repeated, and missing observations:
The original dataset was checked before modeling. It contains 299 rows and 13 columns (12 features and the target), with no missing values and no duplicate rows, so these checks were confirmatory and no imputation or removal was needed. The artificial records will be checked for missing values. Repeated or near-identical records are expected in generated data, so they will not be removed automatically. Instead, we will confirm that no record appears in both the training and test sets. Because artificial records are generated from training data only, they cannot copy test patients.

An empirical description of the relevant distributions:
The distributions of the clinical variables will be examined using descriptive statistics and visualizations. Continuous variables such as age, creatinine phosphokinase, ejection fraction, platelets, serum creatinine, serum sodium, and follow-up time will be summarized using measures such as the mean, median, minimum, maximum, and standard deviation. Binary variables such as anemia, diabetes, high blood pressure, sex, smoking, and death event will be summarized using counts and percentages. In the original data, 203 patients survived (67.9%) and 96 died (32.1%), so the target is imbalanced. Follow-up time ranges from 4 to 285 days (mean 130.3, median 115). Stratified splits and class weights will be used to handle the imbalance.

An appropriate baseline:
A majority-class classifier that predicts "survived" for every patient will be the baseline. In repeated stratified cross-validation on the training set it achieved an accuracy of 0.678 ± 0.009. It detects none of the deaths (recall of 0), and as a constant predictor its ROC-AUC is 0.50. The Logistic Regression, Random Forest, and XGBoost models should outperform this baseline to demonstrate that they provide useful predictive information. The 80% accuracy goal is about 12 points above the baseline. Because accuracy alone hides how many deaths a model misses, all three models will also be compared with the baseline on recall, F1-score, and ROC-AUC.

An initial ranking, regression, classification, or other analysis:
This project is a binary classification problem. The features are the 12 clinical variables, and the target is death during follow-up. Logistic Regression uses standardized features, and class weights are used in all models to handle the class imbalance. The variable time (follow-up days) ends when a patient dies or is censored, so it would not be known when predicting for a new patient and may leak information about the outcome. Each model is therefore evaluated twice, with and without time.

Preliminary results (repeated stratified 5-fold cross-validation with 10 repeats, training set only, no tuning):

Features	Model	Accuracy	Recall	F1	ROC-AUC
with time	Logistic Regression	0.805	0.782	0.720	0.882
with time	Random Forest	0.842	0.777	0.760	0.916
with time	XGBoost	0.837	0.724	0.740	0.912
without time	Logistic Regression	0.741	0.710	0.637	0.786
without time	Random Forest	0.747	0.614	0.607	0.775
without time	XGBoost	0.726	0.598	0.582	0.742

Removing time lowered accuracy by about 6 to 11 points and ROC-AUC by about 10 to 17 points, which is consistent with the leakage concern. Feature importance (coefficients for Logistic Regression, importance scores for Random Forest and XGBoost) will be examined to see which variables drive predictions.

A train–validation–test design or another appropriate evaluation protocol:
The real data will be split once with a stratified 80/20 split (239 training patients, 60 test patients). Because a single validation set drawn from 299 patients gives a noisy estimate, model comparison and tuning will use repeated stratified 5-fold cross-validation (10 repeats) on the training set only, which replaces a fixed validation set. Feature scaling is done inside each fold so that no information leaks across folds. Hyperparameters will be tuned by cross-validation on the training set only, using **[tuning method and hyperparameter ranges]**. The test set will be kept separate from the training and tuning process and used once, at the end, for the final evaluation of each model. Accuracy will be the primary metric because the project has an 80% accuracy goal, and precision, recall, F1-score, and ROC-AUC will be reported alongside it, with the mean and standard deviation across folds.

A distinction between supported claims and limitations:
The project can support claims about how well Logistic Regression, Random Forest, and XGBoost predict death during follow-up within the available dataset and which of the three models performs best under the selected evaluation procedure. In preliminary cross-validation, the 80% accuracy goal was reached only when follow-up time was included (best accuracy 0.842). Without it, the best accuracy was 0.747. Because follow-up time would not be available when predicting for a new patient, the version without it is the more realistic estimate, and the with-time results should be read as an upper bound.

However, the results cannot establish that the models will perform equally well on all patients with heart failure or in a real clinical environment. The original dataset contains only 299 patients from one study, which limits the generalizability of the results and makes scores noisy.

Additionally, the 1,000 artificial records are generated from the training data and should not be considered equivalent to 1,000 new independent patients, so any benefit from them will be judged only on held-out real patients. Therefore, the models' performance should be interpreted as an initial predictive analysis rather than evidence that the models are ready for clinical use.

Dataset:

UCI Heart Failure Clinical Records (Chicco, D., & Jurman, G. (2020). Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making*, 20, 16.)

299 instances of data

Twelve (12) clinical features plus one target:

- age: age of the patient (years)
- anaemia: decrease of red blood cells or hemoglobin (boolean)
- creatinine phosphokinase (CPK): level of the CPK enzyme in the blood (mcg/L)
- diabetes: if the patient has diabetes (boolean)
- ejection fraction: percentage of blood leaving the heart at each contraction (percentage)
- high blood pressure: if the patient has hypertension (boolean)
- platelets: platelets in the blood (kiloplatelets/mL)
- sex: woman or man (binary)
- serum creatinine: level of serum creatinine in the blood (mg/dL)
- serum sodium: level of serum sodium in the blood (mEq/L)
- smoking: if the patient smokes or not (boolean)
- time: follow-up period (days). It ends at death or censoring and is not available at prediction time, so it is evaluated both included and excluded.
- [target] death event: if the patient died during the follow-up period (boolean)